

# EMMANUEL OYELANA

AI Governance & Model Risk • Trustworthy AI for Financial Services

Data & AI Engineer • AI Evaluation, Validation & Governance

- |                                                                                                     |                                                                                                                   |
|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
|  German citizen      |  EU/EFTA - Permit B              |
|  Zurich, Switzerland |  oyelanaemmanuel@rocketmail.com  |
|  eoyelana.com        |  linkedin.com/in/emmanueloyelana |

## PROFESSIONAL SUMMARY

Data and AI engineer with 6+ years of experience building and validating data and machine learning systems in highly regulated industries, now specialising in AI governance and model risk for financial services. Combines hands-on evaluation of large language models (LLMs), covering truthfulness, robustness, bias and instruction following, with regulated industry validation practice across data quality frameworks, independent verification, model documentation and audit readiness under GxP, ICH-GCP and 21 CFR Part 11. This experience maps directly to the model risk, explainability and governance expectations of FINMA Guidance 08/2024 and the EU AI Act. Team winner of the RiskOn Hackathon 2025 sponsored by Julius Baer. German citizen with EU/EFTA work authorisation, based in Zurich. Fluent in German and English with native-level Swiss German comprehension.

## CORE COMPETENCIES

- **AI Governance & Model Risk:** Model validation and evaluation, explainability, bias and robustness testing, model monitoring, documentation and audit readiness, alignment with FINMA 08/2024 and the EU AI Act, three lines of defence.
- **Generative AI & ML:** LLM fine-tuning and evaluation, NLP, agentic AI, model deployment, Anthropic Claude and OpenAI APIs, predictive analytics, deep learning, feature engineering, MLOps.
- **Data Engineering:** ETL and ELT pipelines, dbt, Airflow, Apache Spark, Databricks, Azure, GCP, AWS, data modelling, data warehousing, API integration, GPU-accelerated data processing.
- **Data Governance & Quality:** Validation plans, data quality frameworks, controls and lineage, translation of GxP, ICH-GCP, 21 CFR Part 11 and CDISC SDTM requirements into technical solutions.
- **Programming & BI:** Python, SQL (PostgreSQL, MySQL, Oracle), PL/SQL, R, Power BI, Tableau, Streamlit.
- **Tools & Platforms:** Git, GitHub, GitLab, Docker, Kubernetes, Jira, Confluence.

## PROFESSIONAL EXPERIENCE

### Business Intelligence Consultant (Freelance) | Nexxiot AG | Zurich, Switzerland

Feb 2026 – Present

- Build end-to-end BI and data solutions for an enterprise IoT and logistics business, owning the full pipeline from SQL extraction to governed reporting.
- Develop ETL pipelines that ingest, clean and standardise data from APIs, databases and enterprise systems, and design efficient data models and schemas.
- Write advanced SQL and DAX to translate business requirements into validated KPIs, and advise stakeholders on data governance standards and quality controls.

### Data Migration Specialist (Contract) | GZO Spital Wetzikon | Zurich, Switzerland

Jan 2026 – Apr 2026

- Designed and automated Python and SQL extraction and transformation workflows to migrate patient records to a new Clinical Information System under strict healthcare compliance and data integrity requirements.
- Engineered reconciliation logic across legacy and Navision systems to detect duplicates and ensure accurate, auditable records.
- Performed rule-based validation, verification and post-migration quality checks, producing stakeholder-ready overviews and audit-ready documentation.

## AI Trainer & Model Evaluator (Freelance) | Outlier | Remote (Zurich, Switzerland)

Jan 2025 – Sep 2025

- Evaluated and validated large language model outputs against rule-based guidelines for truthfulness, robustness, bias, conciseness and strict instruction following.
- Authored evaluation documentation and refined evaluation frameworks, improving the consistency, explainability and reproducibility of reviews.
- Built and optimised scalable data ingestion and ETL workflows in Databricks and Azure to process multilingual German and English NLP datasets.

## Lead eClinical Programmer & Data Scientist | Metronomia Clinical Research GmbH | Munich, Germany

Sep 2019 – Dec 2024

*Earlier roles at Metronomia: Clinical Data Manager & EDC Developer (2020–2022), Validation Manager (2020), Data Management Intern (2019–2020).*

- Translated regulatory and data governance requirements (GxP, ICH-GCP, 21 CFR Part 11, CDISC SDTM) into automated, validated technical solutions.
- Engineered and optimised end-to-end ETL pipelines and data models in SQL and Python, improving processing speed and reliability by around 30%.
- Built a data quality monitoring framework and authored Data Validation Plans and SOPs, embedding validation logic and controls across SQL workflows for audit readiness.
- Conducted independent code reviews and established development best practices to ensure high quality, compliant and traceable code.
- Developed dashboards in SQL, Power BI and Python to track data quality metrics, standardising processes and cutting project turnaround by around 20%.

## Intern, Pharmaceutical Analytical Services | Nuvisan GmbH | Neu-Ulm, Germany

2017 – 2019

- Supported pharmaceutical quality assurance and regulatory compliance reporting as well as on-site clinical data entry.

## PUBLICATIONS & AWARDS

- **Team Winner, RiskOn Hackathon 2025, sponsored by Julius Baer, Switzerland.** Winning project "Quality Assurance for Client Contact Notes: How can AI help?" applied NLP, LLMs and text analytics to risk management and compliance in banking.
- **Advancing Risk Management in Swiss Banking: AI-driven Client Documentation, Identity Matching and Fraud Detection.** University of Zurich, Department of Finance, 2026. Examines AI-enabled approaches to KYC, identity matching and fraud detection in Swiss banking.
- **Talent and Workforce Development for AI: Building AI-Related Skills Among Frontline Workers and Managers.** University of Zurich, Faculty of Business, Economics and Informatics, 2026.

## EDUCATION

### M.Sc. Data Science | University of Zurich, Switzerland (Expected Feb 2028)

Coursework: Advanced Data Engineering, Advanced Machine Learning, Cloud Architecture, Data Governance.

### B.Sc. Data Science in Medicine | Technische Hochschule Ulm, Germany (2020)

Thesis: Agile development and validation of a web-based Medical Coding Tool.

## CERTIFICATIONS

- NVIDIA-Certified Professional: Accelerated Data Science
- Microsoft Certified: Azure AI Fundamentals, Azure Data Fundamentals
- Databricks Certified: Fundamentals, Data Engineer, Apache Spark Developer
- dbt Fundamentals, GitHub Foundations
- Medidata Rave EDC: Study Builder, Principal Investigator

## LANGUAGES

- German: fluent, with native-level Swiss German comprehension • English: fluent